

Package ‘manMetaVAR’

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Compress	<i>Compress Replication</i>
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Description

Compress Replication

Usage

Compress(taskid, repid, output_folder)

Arguments

- | | |
|---------------|-----------------------------------|
| taskid | Positive integer. Task ID. |
| repid | Positive integer. Replication ID. |
| output_folder | Character string. Output folder. |

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

FitDTVAR

Fit the Model using the fitVARMxID Package

Description

The function fits the model using the [fitVARMxID::fitVARMxID](#) package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

data	R object. Output of the GenData() function.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

Description

The function performs multivariate meta-analysis using the [metaDyn::metaDyn](#) package.

Usage

```
FitMetaVAR(fit, ncores = NULL, seed = NULL)
```

Arguments

<code>fit</code>	R object. Output of the FitDTVAR() function.
<code>ncores</code>	Positive integer. Number of cores to use.
<code>seed</code>	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
fit <- FitDTVAR(data = data, seed = seed)
meta <- FitMetaVAR(fit = fit, seed = seed)
summary(meta)
print(meta)
coef(meta)
vcov(meta)

## End(Not run)
```

Description

The function fits the model using Mplus.

Usage

```
FitMplus(  
  data,  
  chains = 2L,  
  iter = 40000L,  
  fscores = NULL,  
  plot = FALSE,  
  wd = ".",  
  mplus_bin = NULL,  
  ncores = NULL,  
  seed = NULL  
)
```

Arguments

data	R object. Output of the GenData() function.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
wd	Character string. Working directory.
mplus_bin	Character string. Path to Mplus binary. If mplus_bin = NULL, the function will try to find the appropriate binary.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:  
seed <- 42  
data <- GenData(taskid = 1, seed = seed)  
fit <- FitMplus(data = data)  
print(fit)  
summary(fit)  
  
## End(Not run)
```

GenData

Simulate Data

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid, seed = NULL)
```

Arguments

taskid	Positive integer. Task ID.
seed	Integer. Random seed.

Examples

```
## Not run:
seed <- 42
data <- GenData(taskid = 1, seed = seed)
print(data)
summary(data)
plot(data)

## End(Not run)
```

manmetavar-data-methods

Methods for Objects of Class manmetavar.data

Description

This page documents the available methods for objects of class `manmetavar.data`.

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

x	Object of class <code>manmetavar.data</code> .
...	additional arguments.
object	Object of class <code>manmetavar.data</code> .

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-dtvar-methods

Methods for Objects of Class `manmetavar.dtvvar`

Description

This page documents the available methods for objects of class `manmetavar.dtvvar`.

Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  robust = FALSE,
  ncores = NULL,
  ...
)
```

```
## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)

## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  mu = TRUE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  digits = 4,
  ncores = NULL,
  ...
)
```

Arguments

object	Object of class <code>manmetavar.dtvar</code> .
mu	Logical. If <code>mu = TRUE</code> , include estimates of the <code>mu</code> vector, if available. If <code>mu = FALSE</code> , exclude estimates of the <code>mu</code> vector.
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the <code>alpha</code> vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the <code>alpha</code> vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the <code>beta</code> matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the <code>beta</code> matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the <code>nu</code> vector, if available. If <code>nu = FALSE</code> , exclude estimates of the <code>nu</code> vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the <code>psi</code> matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the <code>psi</code> matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the <code>theta</code> matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the <code>theta</code> matrix.
ncores	Positive integer. Number of cores to use.

...	additional arguments.
robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix.
x	Object of class <code>manmetavar.dtvvar</code> .
means	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
digits	Integer indicating the number of decimal places to display.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-metavar-methods

Methods for Objects of Class `manmetavar.metavar`

Description

This page documents the available methods for objects of class `manmetavar.metavar`.

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.metavar'
vcov(object, robust = NULL, ...)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = NULL, robust = NULL, digits = 4, ...)

## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, robust = NULL, ...)
```

Arguments

object	Object of class <code>manmetavar.metavar</code> .
...	additional arguments.
robust	Logical. If TRUE, use robust (sandwich) sampling variance-covariance matrix. If FALSE, use normal theory sampling variance-covariance matrix. If NULL, the function will check object if robust standard errors are available.
x	Object of class <code>manmetavar.metavar</code> .

alpha	Numeric vector. Significance level α .
digits	Integer indicating the number of decimal places to display.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.

Author(s)

Ivan Jacob Agaloos Pesigan

manmetavar-mplus-methods

Methods for Objects of Class manmetavar.mplus

Description

This page documents the available methods for objects of class `manmetavar.mplus`.

Usage

```
## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)

## S3 method for class 'manmetavar.mplus'
plot(
  x,
  what = "posterior",
  parm = NULL,
  level = 0.95,
  burnin = NULL,
  legend_loc = "topright",
  ...
)
```

Arguments

<code>object</code>	Object of class <code>manmetavar.mplus</code> .
<code>median</code>	Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.
<code>burnin</code>	Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.
<code>...</code>	additional arguments.
<code>alpha</code>	Numeric vector. Significance level α .
<code>digits</code>	Integer indicating the number of decimal places to display.
<code>parm</code>	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
<code>level</code>	the confidence level required.
<code>x</code>	Object of class <code>manmetavar.mplus</code> .
<code>what</code>	Character string. If <code>what = "posterior"</code> , return posterior distribution plots. If <code>what = "trace"</code> , return trace plots.
<code>legend_loc</code>	Character string. Legend location.

Author(s)

Ivan Jacob Agaloos Pesigan

model	<i>Model Parameters</i>
-------	-------------------------

Description

Model Parameters

Usage

```
data(model)
```

Format

A list with 15 elements:

k Number of variables.

mu_mu Mean of the set-point vector μ .

mu_sigma Covariance matrix of the parameter μ .

mu_sigma_l Cholesky factor of the covariance matrix of the parameter μ .

beta_mu Mean of lagged coefficients matrix β .

beta_sigma Covariance matrix of the parameter vec (β).

beta_sigma_1 Cholesky factor of the covariance matrix of the parameter $\text{vec}(\beta)$.

psi Process noise covariance matrix Ψ .

psi_1 Cholesky factor of the process noise covariance matrix Ψ .

psi_d_ldl uc_d of the LDL' decomposition of the process noise covariance matrix Ψ . See [fitVARMxID::LDL\(\)](#).

psi_l_ldl s_1 of the LDL' decomposition of the process noise covariance matrix Ψ . See [fitVARMxID::LDL\(\)](#).

ma_fixed Vector of fixed effects $\theta = [\mu, \text{vec}(\beta)]'$

ma_random Matrix of random effects $\theta = [\mu, \text{vec}(\beta)]'$

ma_random_d_ldl uc_d of the LDL' decomposition of the random effects. See [fitVARMxID::LDL\(\)](#).

ma_random_l_ldl s_1 of the LDL' decomposition of the random effects. See [fitVARMxID::LDL\(\)](#).

Author(s)

Ivan Jacob Agaloos Pesigan

MplusInput

Create Mplus Input File

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  ncores = NULL,
  seed = NULL
)
```

Arguments

fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.

fcores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#)

Examples

```
cat(
  MplusInput(
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 40000L,
    fcores = 1000L,
    plot = TRUE,
    ncores = NULL,
    seed = 42
  )
)
```

params

Simulation Parameters

Description

Simulation Parameters

Usage

```
data(params)
```

Format

A dataframe with 9 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occasions.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim

Simulation Replication

Description

Simulation Replication

Usage

```

Sim(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  seed,
  data,
  metavar,
  mplus,
  chains,
  iter,
  fscores,
  plot
)

```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
data	Logical. Simulate data.
metavar	Logical. If metavar = TRUE, fit the model using the metaDyn package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR

Simulation Replication - FitDTVAR

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR	<i>Simulation Replication - FitMetaVAR</i>
---------------	--

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus

Simulation Replication - FitMplus

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3;</code> to Mplus input file.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
-------	-----------------------------

Description

Simulation File Name

Usage

SimFN(output_type, output_folder, suffix)

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-mplus".
output_folder	Character string. Output folder.
suffix	Character string. Output of manCTMed:::SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
------------	---

Description

Simulation Replication - GenData

Usage

SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj	<i>Simulation Project Name</i>
---------	--------------------------------

Description

Simulation Project Name

Usage

SimProj()

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

Sum	<i>Summary</i>
-----	----------------

Description

Summary

Usage

```
Sum(  
  taskid,  
  reps,  
  output_folder,  
  overwrite,  
  integrity,  
  metavar_normal,  
  metavar_robust,  
  naive,  
  mplus,  
  ncores  
)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
metavar_normal	Logical. If metavar_normal = TRUE, summarize metaVAR model using normal confidence intervals.
metavar_robust	Logical. If metavar_robust = TRUE, summarize metaVAR model using robust (sandwich) confidence intervals.
naive	Logical. If naive = TRUE, summarize naive estimates.
mplus	Logical. If mplus = TRUE, summarize the DSEM model.
ncores	Positive integer. Number of cores to use.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal	<i>Summary: Normal Theory (FitMetaVAR)</i>
---------------------	--

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARRobust	<i>Summary: Robust (FitMetaVAR)</i>
---------------------	-------------------------------------

Description

Summary: Robust (FitMetaVAR)

Usage

```
SumFitMetaVARRobust(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplus

Summary (FitMplus)

Description

Summary (FitMplus)

Usage

```
SumFitMplus(taskid, reps, output_folder, overwrite, integrity, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitNaive	<i>Summary: Naive (FitMetaVAR)</i>
-------------	------------------------------------

Description

Summary: Naive (FitMetaVAR)

Usage

SumFitNaive(taskid, reps, output_folder, overwrite, integrity, ncores)

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
ncores	Positive integer. Number of cores to use.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

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